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DUNMAN HIGH SCHOOL

Preliminary Examinations

Year 6

H1 CHEMISTRY

Paper 2 Section A
 Section B

8872/02

24 September 2013
2 hours

Additional Materials: Data Booklet & Writing Paper

INSTRUCTIONS TO CANDIDATES

- 1 Answer **all** questions in both sections.
- 2 Write your **name** and **class** on this cover page.

Section A

- 3 Write your answers in the spaces provided on this question paper.
- 4 **You are advised to spend a maximum of 60 minutes on Section A.**

Section B

- 5 Write your **name** and **class** on the Cover Sheet provided.
- 6 Write your answers on the separate writing papers provided.
- 7 **Start each question on a fresh sheet of paper.**
****[Marks will be deducted if you fail to do so.]***
- 8 At the end of the examination:
 - Staple or fasten all your work securely together with the Cover Sheet on top.
 - Hand in the question paper separately.
- 9 **You are advised to spend a maximum of 60 minutes on Section B.**

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question.

You are advised to show all workings in calculations.

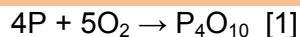
You are reminded of the need for good English and clear presentation in your answers.

For Examiner's Use	
Question	Section A
1	11
2	9
3	11
4	9
Total	40

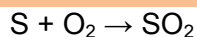
Section A

Answer **all** questions in the spaces provided.

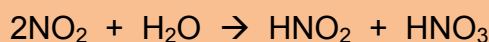
- 1(a) (i)** Write equations to illustrate the formation of
- phosphorous(V) oxide when phosphorus is heated with oxygen



- sulfur(IV) oxide when sulfur is heated with oxygen



- (ii)** When NO_2 reacts with water, nitrogen undergoes a disproportionation reaction to give a mixture of two acids as shown:



Use this reaction to explain the term *disproportionation*.

In the reaction, oxidation state of N increases from +4 in NO_2 to +5 in HNO_3 and decreases to +3 in HNO_2 at the same time.

- (iii)** In a similar disproportionation reaction, ClO_2 reacts with aqueous NaOH to produce a solution containing two chlorine-containing sodium salts. Suggest the identities of the two chlorine-containing sodium salts.

[4]



- (b)** The major source of sulfur for the manufacture of sulfuric acid by the Contact process is the de-sulfurisation of 'sour' natural gas. Many natural gas wells produce a mixture of volatile hydrocarbons (mainly CH_4 and C_2H_6) together with up to 25% hydrogen sulfide, H_2S .

- (i)** Complete and balance the following equation showing the complete combustion of a gaseous mixture consisting of 2 mol of CH_4 , 1 mol of C_2H_6 and 1 mol of H_2S .



- (ii)** Suggest why it is important to remove the H_2S before burning the natural gas industrially.

SO_2 produced causes acid rain.

H_2S is removed by passing the 'sour' natural gas through a solvent containing ethanolamine, $\text{HOCH}_2\text{CH}_2\text{NH}_2$. The following reaction takes place.



- (iii)** State the type of reaction that has occurred.

Acid-base reaction

- (iv) If a sample of natural gas contains 5% by volume of H_2S , calculate the mass of ethanolamine required to remove all the H_2S from a 1000 dm^3 sample of gas, measured under room conditions.

1000 dm^3 contains 50 dm^3 of H_2S

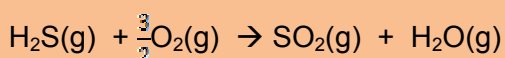
Moles of $\text{H}_2\text{S} = 50/24 = 2.08 \text{ mol}$

Mr (ethanolamine) = $24 + 7 + 14 + 16 = 61.0$

Mass of ethanolamine needed = $2.08 \times 61.0 = 127 \text{ g}$

The recovered H_2S is converted to sulfur by the following two reactions.

- I Part of the H_2S is burned in air.



- II The gas stream resulting from reaction I is then blended with the remaining H_2S and fed into an iron oxide catalyst bed, where sulfur and water are produced according to the equation.



- (v) Use the following data to calculate ΔH^θ for the reaction between H_2S and SO_2 .

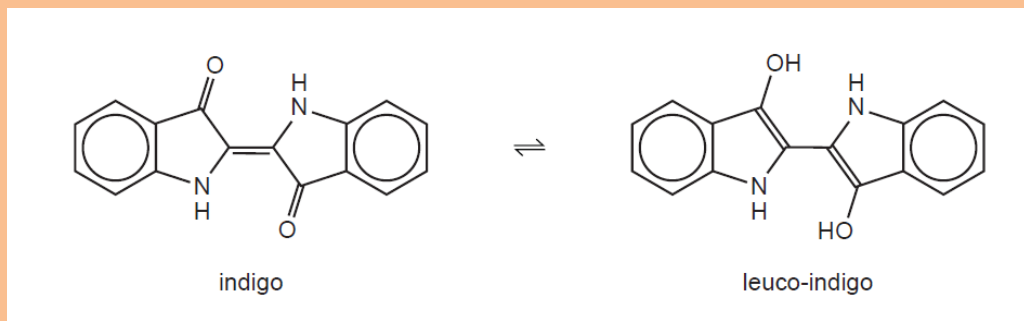
Compound	$\Delta H_f^\theta / \text{kJ mol}^{-1}$
$\text{H}_2\text{S}(\text{g})$	-21
$\text{SO}_2(\text{g})$	-297
$\text{H}_2\text{O}(\text{g})$	-242
$\text{S}(\text{g})$	+11

$$\begin{aligned}\Delta H^\theta &= [3(11) + 2(-242)] - [2(-21) + (-297)] \\ &= -451 + 339 \\ &= -112 \text{ kJmol}^{-1}\end{aligned}$$

[Total: 11]

- 2** Indigo is the dye used in blue jeans. Although originally extracted from plants of the type *indigofera*, it is now almost entirely made artificially.

Indigo is insoluble in water but this disadvantage can be overcome by converting it into the water-soluble colourless Leuco-indigo. If cloth soaked in a solution of Leuco-indigo is left to dry in the air, the Leuco-indigo is converted into the insoluble blue indigo, which is precipitated out onto the fibres of the cloth.



- (a)** On the structures above, circle and name four different functional groups present in either indigo or leuco-indigo, which are not benzene rings.

[3]

Alkene, carbonyl (ketone), amine and alcohol

- (b)** Calculate the percentage by mass of carbon in Indigo.

[2]

Mr of Indigo = 262 ($C_{16}H_{10}N_2O_2$)

Mass of carbon present = $[(16 \times 12)/262] \times 100\% = 73.3\%$

- (c)** **(i)** What *type of reaction* is the conversion of Indigo into Leuco-indigo?

Reduction

- (ii)** Suggest a laboratory reagent for this reaction.

[2]

$LiAlH_4$ in dry ether or $NaBH_4$ (in methanol)
Reject H_2/Ni as an answer.

- (d)** Suggest the reagents and conditions for a reaction that could be used to distinguish between Indigo and Leuco-indigo by identifying Indigo positively. In your test, state the expected observation for each compound.

[2]

Test	Indigo	Leuco-indigo
2,4-DNPH, warm	Orange ppt	No orange ppt

- 3** Sodium benzoate is a preservative and is able to prevent the growth of bacteria and fungi. It is most widely used in acidic foods such as salad dressings, carbonated drinks, jams and fruit juices.

According to the United States Food and Drug Administration, all food preservatives are limited to an amount of 0.1% by weight. However, The International Programme on Chemical Safety found no adverse effects of sodium benzoate in humans at doses of

647–825 mg/kg of body weight per day.

In acidic foods, sodium benzoate is absorbed into the food cells as benzoic acid. This lowers the intracellular pH of food, creating an environment which fungi and bacteria cannot thrive. In typical fruit juices which have a pH of 3.6, the concentration of benzoic acid was found to be $0.001 \text{ mol dm}^{-3}$. In foods with a pH level of 4.2 to 4.5, a mixture of benzoic acid and sodium benzoate is present. In foods above pH 4.5, sodium benzoate becomes ineffective as a preservative.

Sodium benzoate is prepared by adding benzoic acid to a hot concentrated solution of sodium carbonate until effervescence ceases, and all reactants are used up. Appropriate separation techniques can be applied to obtain pure solid sodium benzoate, which is stable upon heating.

An alternative to sodium benzoate is potassium benzoate, which works in exactly the same manner.

- (a) A typical jar of jam weighs 500 g. Calculate its maximum allowable mass of sodium benzoate as a preservative.

Maximum mass of sodium benzoate allowed in 500 g of jam

$$= \frac{0.1}{100} \times 500$$

$$= \mathbf{0.5 \text{ g}}$$

[1]

- (b) Assuming that an average adult weighs 70 kg, calculate the maximum mass of sodium benzoate a person can intake per day.

Maximum mass of sodium benzoate a person can intake

$$= 825 \times 10^{-3} \times 70$$

$$= 57.75 \text{ g}$$

$$= \mathbf{57.8 \text{ g}}$$

[1]

- (c) (i) Calculate the concentration of H^+ ions in typical fruit juices.

$$\text{pH of fruit juices} = 10^{-3.6} = \mathbf{2.51 \times 10^{-4} \text{ mol dm}^{-3}}$$

- (ii) Hence, deduce the value of K_a of benzoic acid, assuming benzoic acid is the only acid present.

$$K_a = \frac{[\text{H}^+][\text{benzoate}]}{[\text{benzoic acid}]}$$

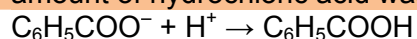
$$= \frac{(10^{-3.6})^2}{0.001} = \mathbf{6.31 \times 10^{-5} \text{ mol dm}^{-3}}$$

[2]

- (d) (i) State the type of solution formed by the mixture of benzoic acid and sodium benzoate in pH 4.2 to 4.5.

The solution is a **buffer** solution.

- (ii) With the aid of a suitable equation, explain what happens when a small amount of hydrochloric acid was added to this mixture.



When a small amount of hydrochloric acid was added, the H^+ was **removed as benzoic acid**. The **pH remained fairly constant**.

[3]

- (e) (i) Write the balanced equation for the formation of sodium benzoate from sodium carbonate.



- (ii) Suggest a simple method to isolate solid sodium benzoate from the reaction in (e) (i).

Pure solid sodium benzoate can be obtained by **evaporating the mixture to dryness**.

[2]

- (f) Predict, with explanations, if potassium benzoate has a higher or lower boiling point than sodium benzoate.

Charge of cation: $\text{K}^+ = \text{Na}^+$

Size of cation: $\text{K}^+ > \text{Na}^+$

Lattice energy: $\text{K}^+ < \text{Na}^+$

Thus, **boiling point: $\text{K}^+ < \text{Na}^+$**

[2]

[Total: 11]

- 4 (a) KMnO_4 and $\text{K}_2\text{Cr}_2\text{O}_7$ are oxidising agents, usually used in acidic solutions.

- (i) State the colour changes you would expect to observe, when each of these reagents are heated with butan-1-ol.

KMnO_4 :	Purple	to	Colourless
$\text{K}_2\text{Cr}_2\text{O}_7$:	Orange	to	Green

- (ii) When heated with butan-1-ol, each of these reagents give rise to different products when different conditions are applied. Suggest how the two products can be obtained by using KMnO_4 and $\text{K}_2\text{Cr}_2\text{O}_7$. Include a balanced equation for each set of reagents and conditions.

Reagents and conditions : KMnO_4 , H_2SO_4 (aq), reflux

Equation : $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH} + 2[\text{O}] \rightarrow \text{CH}_3\text{CH}_2\text{CH}_2\text{COOH} + \text{H}_2\text{O}$

Reagents and conditions : $\text{K}_2\text{Cr}_2\text{O}_7$, H_2SO_4 (aq), heat with distillation

Equation : $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH} + [\text{O}] \rightarrow \text{CH}_3\text{CH}_2\text{CH}_2\text{CHO} + \text{H}_2\text{O}$

[6]

- (b) The following table shows the solubilities of butan-1-ol and but-1-ene in water.

Compound	Mass soluble in 1 dm ³
Butan-1-ol	73.0
But-1-ene	2.21

Explain the above data.

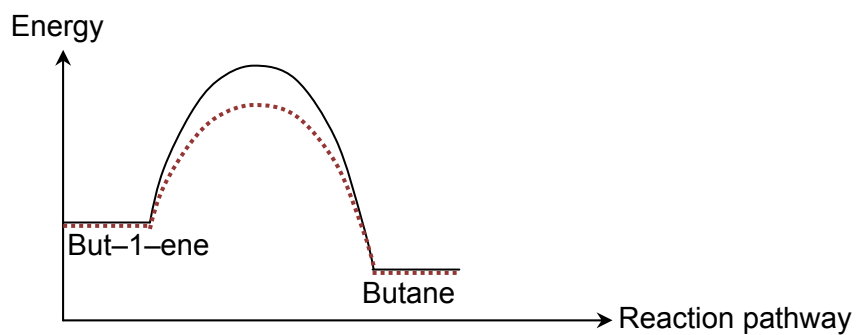
Butan-1-ol is soluble (or more soluble) in water as favourable **hydrogen bonds** can be formed between butan-1-ol and water molecules.

But-1-ene is not soluble (or has low solubility) in water as the **weak Van der**

Waals' forces between the but-1-ene molecules **cannot overcome the stronger hydrogen bonds** between the water molecules.

[2]

- (c) The energy profile diagram below represents the hydrogenation of but-1-ene to butane. On the same diagram, draw the energy profile when nickel is added to the reaction.

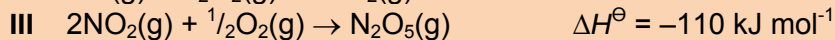
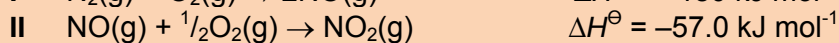


[1]

[Total: 9]

Answer any **two** questions.

- 5** Dinitrogen pentoxide, N_2O_5 , can be produced by the following reaction sequence in the car engine:



- (a) (i)** By using relevant data in the *Data Booklet*, calculate the bond energy of the N-O bond in nitric oxide in reaction I.

$$\text{BE}(\text{N}_2) + \text{BE}(\text{O}_2) - 2\text{BE}(\text{NO}) = 180$$

$$994 + 496 - 2\text{BE}(\text{NO}) = 180$$

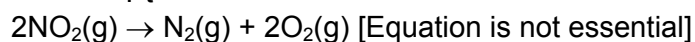
$$\text{BE}(\text{NO}) = +655 \text{ kJ mol}^{-1}$$

- (ii)** Suggest an explanation why reaction I occurs in car engine.

The high temperature in the car engine is able to overcome the strong $\text{O}=\text{O}$ bond and $\text{N}\equiv\text{N}$ bond.

- (iii)** Nitrogen dioxide is a pollutant that is often produced in car engine. Explain how the pollutant can be removed in the car engine.

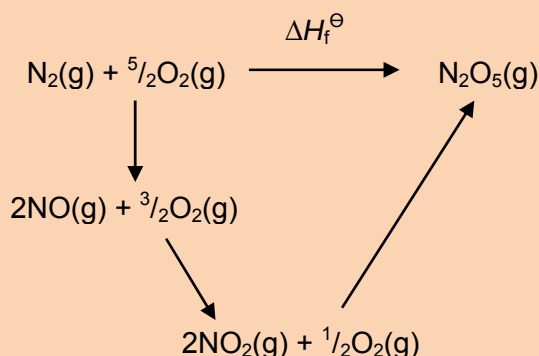
Pt



The nitrogen dioxide is converted to nitrogen and oxygen gas in the catalytic convertor (with Pt as catalyst) in the car engine.

Pt catalyst is not the essential marking point. As long as catalyst/catalytic convertor is mentioned.

- (iv)** By using the given energy cycle and the data above, calculate the standard enthalpy change of formation of dinitrogen pentoxide. [2]



By Hess' law,

$$\begin{aligned} \Delta H_f^\ominus(\text{N}_2\text{O}_5) &= +180 + 2(-57.0) - 110 \\ &= -44.0 \text{ kJ mol}^{-1} \end{aligned}$$

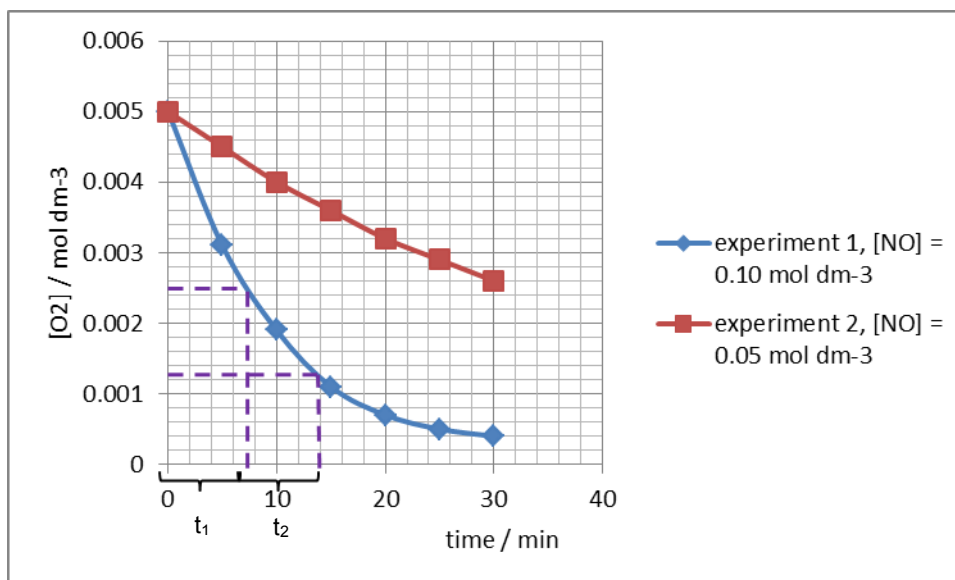
- (b)** The rate of reaction for reaction II was investigated and the following data was obtained.

Time/minutes	Experiment 1, with $[\text{NO}] = 0.10 \text{ mol dm}^{-3}$	Experiment 2, with $[\text{NO}] = 0.05 \text{ mol dm}^{-3}$
	$[\text{O}_2] / \text{mol dm}^{-3}$	$[\text{O}_2] / \text{mol dm}^{-3}$
0	0.0050	0.0050
5	0.0031	0.0045
10	0.0019	0.0040
15	0.0011	0.0036
20	0.0007	0.0032
25	0.0005	0.0029
30	0.0004	0.0026

- (i) Explain why nitrogen monoxide is used in large excess.

This is to ensure that the concentration of nitrogen monoxide remains approximately constant so that the rate of reaction will be independent of the concentration of nitrogen monoxide.

- (ii) Using the same axes, plot graphs of $[\text{O}_2]$ against time for the two experiments.



- (iii) Use your graphs to determine the order of reaction with respect to O_2 and NO , showing your workings clearly.

Using the curve for experiment 1, $t_1 \approx t_2 \approx 7 \text{ min}$ (accept 6.5 – 7 min). The half-lives of O_2 are approximately constant at 7 min, hence the reaction is first order with respect to O_2 .

When $[\text{NO}] = 0.1 \text{ mol dm}^{-3}$, initial rate = $4 \times 10^{-4} \text{ mol dm}^{-3} \text{ min}^{-1}$

When $[\text{NO}] = 0.05 \text{ mol dm}^{-3}$, initial rate = $1 \times 10^{-4} \text{ mol dm}^{-3} \text{ min}^{-1}$

When $[\text{NO}]$ is doubled, the initial rate is quadrupled. Hence, the reaction is second order with respect to NO .

- (iv) State the rate equation of reaction II and hence calculate the rate constant, including its units.

$$\text{rate} = k[\text{O}_2][\text{NO}]^2$$

Using initial rate method:

$$4 \times 10^{-4} = k(0.005)(0.1)^2$$

$$k = 8.00 \text{ mol}^{-2} \text{ dm}^6 \text{ min}^{-1}$$

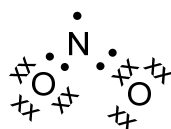
or using half-life method:

$$\text{rate} = k'[\text{O}_2] \text{ where } k' = k[\text{NO}]^2$$

$$k' = \ln 2 / 7 = 0.09902 \text{ min}^{-1}$$

$$k = 0.09902 / (0.1)^2 = 9.90 \text{ mol}^{-2} \text{ dm}^6 \text{ min}^{-1}$$

(c)	The nitrogen dioxide produced in reaction II is able to form dinitrogen tetraoxide as shown below: $2\text{NO}_2(\text{g}) \rightleftharpoons \text{N}_2\text{O}_4(\text{g})$ brown colourless
(i)	Draw a dot-and-cross diagram to illustrate the bonding in a nitrogen dioxide molecule and predict its shape. [2]



shape about nitrogen atom = bent

(ii)	Explain whether the enthalpy change of dimerisation is endothermic or exothermic. Hence, predict the colour change of the reaction mixture when the temperature is increased. [2]
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The forward reaction is exothermic as N-N bond is formed during dimerisation. By Le Chatelier's principle, an increase in temperature favours the backward reaction to absorb heat (or the position of equilibrium shifts to the left). Hence, the colour of the reaction mixture becomes darker brown.

[Total: 20]

6 (a)	An experiment is conducted to determine the value of n in the formula of a cation $[\text{Cu}(\text{NH}_3)_n]^{2+}$. This cation is formed when copper(II) reacts with ammonia. 100 cm³ of 0.100 mol dm⁻³ of copper(II) sulfate is mixed with 100 cm³ of aqueous ammonia. The resulting mixture, which contains the cation $[\text{Cu}(\text{NH}_3)_n]^{2+}$ and an excess of ammonia, is then shaken with trichloromethane and allowed to stand for equilibrium to be established. The excess ammonia is dissolved in both the aqueous layer and the trichloromethane layer. 25.0 cm³ of trichloromethane layer is found to neutralise 26.00 cm³ of 0.025 mol dm⁻³ hydrochloric acid while 25.0 cm³ of aqueous layer required 21.00 cm³ of 1.00 mol dm⁻³ hydrochloric acid for neutralisation. The equilibrium constant of ammonia between the aqueous layer and the trichloromethane layer is 25.0. $\text{NH}_3(\text{aq}) \rightleftharpoons \text{NH}_3(\text{CHCl}_3)$
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$$K = \frac{[\text{ammonia}]_{\text{aqueous}}}{[\text{ammonia}]_{\text{CHCl}_3}}$$

Calculate:

- (i) the concentration of ammonia in the trichloromethane layer.

$$[\text{NH}_3] = 26.00/25.0 \times 0.025 = \mathbf{0.026 \text{ mol dm}^{-3}}$$

- (ii) the concentration of free ammonia in the aqueous layer, using the equilibrium constant.

$$K_{\text{Partition coefficient}} = \frac{[\text{ammonia}]_{\text{aqueous}}}{[\text{ammonia}]_{\text{CHCl}_3}} = 25.0$$

$$[\text{ammonia}]_{\text{aqueous}} = 0.026 \times 25.0 = \mathbf{0.65 \text{ mol dm}^{-3}}$$

- (iii) the total concentration of ammonia (free and in the cation $[\text{Cu}(\text{NH}_3)_n]^{2+}$) in the aqueous layer.

$$\text{Total } [\text{ammonia}] = 21.00/25.0 \times 1.00 = \mathbf{0.84 \text{ mol dm}^{-3}}$$

- (iv) the value of n in the formula of the cation $[\text{Cu}(\text{NH}_3)_n]^{2+}$, stating one assumption made in the calculation.

[7]

$$[\text{ammonia}]_{\text{(complexed)}} = 0.84 - 0.65 = \mathbf{0.19 \text{ mol dm}^{-3}}$$

$$n_{\text{Cu}^{2+}} = 0.100 \times 0.100 = 0.0100 \text{ mol}$$

$$n_{\text{ammonia (complexed)}} = 0.19 \times 200/1000 = 0.038 \text{ mol}$$

$$n = 0.038/0.01 = 3.8 \\ \approx \mathbf{4}$$

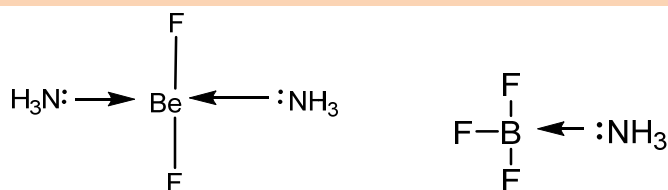
Assumption:

1. All the ammonia in the compound, $[\text{Cu}(\text{NH}_3)_n]^{2+}$, reacted with hydrochloric acid.
2. None of the copper complex ion is partitioned/ migrated to the organic layer.

- (b) Both beryllium difluoride and boron trifluoride can also react with ammonia. However, their mole ratios are different.

Beryllium difluoride and ammonia react in a 1:2 ratio while boron trifluoride and ammonia react in a 1:1 ratio.

- (i) Draw diagrams to show the bonding within these molecules and state the type of bond formed in the products.



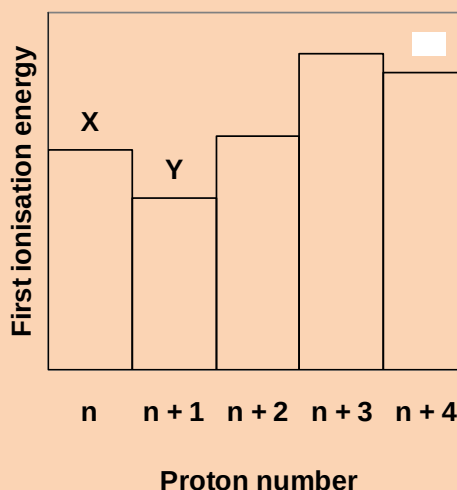
BeF_2 and BF_3 react with NH_3 through **dative (or co-ordinate) bonds**.

- (ii) Explain why the mole ratios are different for these two compounds.

[6]

Beryllium in BeF_2 has only 4 valence electrons while boron in BF_3 has 6 valence electrons. Hence beryllium in BeF_2 needs to form two dative bonds with two ammonia molecules while boron in BF_3 only needs to form one dative bond with one ammonia molecule, in order to achieve stable octet electronic configuration.

- (c) The graph below shows the variation in first ionisation energy of some elements in Period 3.



Elements X and Y have proton numbers n and $n+1$ respectively.

- (i) By considering the trend in first ionisation energy, state and explain which Group element Y belongs to.

Group III. A Group III metal has a **lower first ionisation energy** than a **Group II element X** because **less energy is required to remove a 3p electron in Y than a 3s electron in X** since a **3p electron is higher in energy** and is more strongly held by the nucleus.

- (ii) Describe the behaviour of the oxide of X in dilute hydrochloric acid and sodium hydroxide respectively.

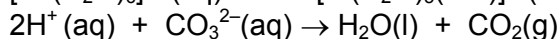
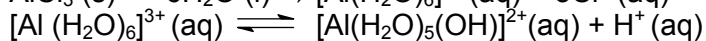
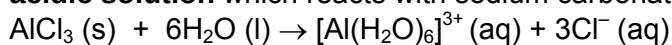
A Group II metal (X) oxide is basic and undergoes **neutralisation with acid** but **does not react with dilute NaOH**.

- (iii) Sodium carbonate was added to a solution of the chloride of Y and effervescence was observed. Explain this observation with the aid of relevant equations.

[7]

[Total: 20]

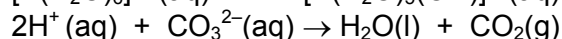
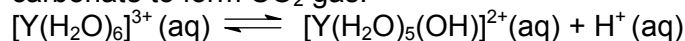
Element Y is in Group III and is Al . AlCl_3 undergoes **hydrolysis** to give an **acidic solution** which reacts with sodium carbonate to form CO_2 gas.



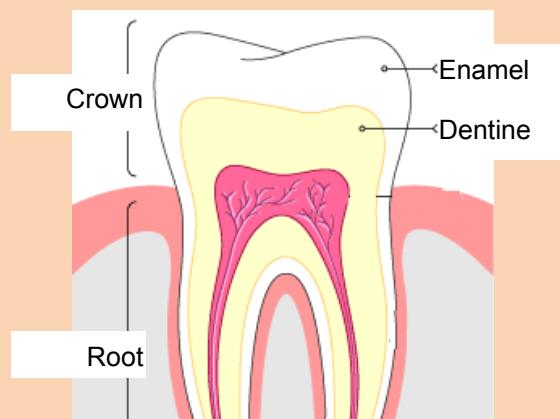
OR

YCl_3 undergoes **hydrolysis** to give an **acidic solution** which reacts with sodium

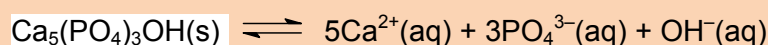
carbonate to form CO₂ gas.



- 3 (a)** A tooth is made up of two parts: the crown and the root. Dental crown is the visible part of the teeth which is made of enamel and dentine.

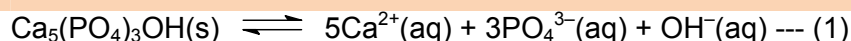


The enamel is made of hydroxyapatite, Ca₅(PO₄)₃OH. In the mouth, mineral substances such as calcium ions and phosphate ions that are present in saliva contribute to the formation and decomposition of the hydroxyapatite. These two processes occur simultaneously until an equilibrium is reached. The formation process is called mineralisation of the enamel, whereas the decomposition process is called demineralisation.



Explain how an acidic medium can affect the demineralisation of teeth.

[2]



In an acidic medium, the **H⁺ ions can react with OH⁻ or PO₄³⁻ ions** produced during demineralisation of hydroxyapatite in eqm (1). By Le Chatelier's Principle, this shifts the equilibrium (1) **forward** (or to the right), **encouraging/ accelerating** the demineralisation process which resulted in tooth decalcification.

- (b)** An organic compound **R** (*M_r* = 113.6) contains by mass 63.2% carbon, 8.7% hydrogen, and 28.1% oxygen. **R** undergoes an addition reaction with bromine in a 1:1 mole ratio. It also reacts with sodium hydrogen carbonate, liberating carbon dioxide gas.

- (i)** Calculate the empirical formula of **R**.

Element	C	H	O
% by mass	63.2	8.7	28.1
No. of moles	63.2 / 12.0 = 5.27	8.7 / 1.0 = 8.7	28.1 / 16.0 = 1.76
Mole ratio	3	5	1

Empirical formula of **R** is **C₃H₅O**.

(ii) Determine the molecular formula of **R**.

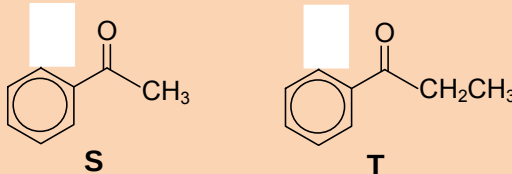
Let the molecular formula of **R** be $(C_3H_5O)_n$.
 $(3 \times 12.0 + 5 \times 1.0 + 16.0)n = 113.6$
 $n = 2$
 Molecular formula of **R** is **C₆H₁₀O₂**.

(iii) Deduce with reasoning, a possible structure for compound **R**.

R contains a $-COOH$ functional group since it undergoes acid–base/neutralisation reaction with sodium hydrogen carbonate gives CO_2 .
R contains **one C=C double bond** as it reacts with bromine in a 1:1 mole ratio.
 Therefore **R** is **CH₂=CHCH(CH₂CH₃)CO₂H**

[8]

(c) Suggest a suitable chemical test to distinguish between molecules **S** and **T** and describe the expected observations for each molecule.



Add **aqueous iodine in aqueous NaOH** to separate samples of **S** and **T**, and **warm**.

[1] S gives a **yellow precipitate** while **T** does **not** give a yellow precipitate.

(d) The elements of Group VII, especially chlorine, have played an important role in the development of chemistry. The following describes a reaction involving a chlorine containing compound.

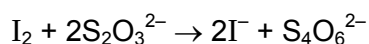
An interhalogen compound **A**, with the formula I_xCl_y , was dissolved in aqueous potassium iodide, and a brown solution was obtained.



The brown solution was titrated with 0.50 mol dm^{-3} sodium thiosulfate and 34.0 cm^3 was required to reach the end-point. Careful addition of aqueous silver nitrate produced a white precipitate of silver chloride, which weighed 1.83 g when dried.

[4]

(i) Calculate the amount of iodine produced in the reaction shown above.



$$\begin{aligned} n(I_2) \text{ produced when reacted with KI} \\ = [0.50 \times (34.0/1000)] / 2 = 8.50 \times 10^{-3} \text{ mol} \end{aligned}$$

(ii) Calculate the amount of chloride ions produced in the reaction shown above.

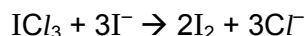
$$n(\text{Cl}^-) = 1.83 / 143.5 = 0.0128 \text{ mol}$$

- (iii) Determine the formula of compound **A** and hence construct a balanced equation between compound **A** and potassium iodide.

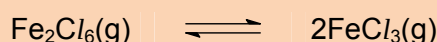
$$n(\text{Cl}^-) : n(\text{I}_2) = 0.0128 : 8.50 \times 10^{-3} = 3 : 2$$

$\Rightarrow$ simplest mole ratio, x:y = 1:3

Hence formula of **A** is ICl_3 .



- (e) Anhydrous iron(III) chloride is a dark brown moisture sensitive solid (melting point = 306°C) which forms an acidic solution in water. It exists as a dimer, Fe_2Cl_6 in the solid state. In the gaseous state, the dimer increasingly dissociates into its monomer until dissociation is complete at 400°C .



0.450 g of Fe_2Cl_6 was placed in a 0.500 dm^3 evacuated flask and heated to 350°C . The amount of gaseous mixture at equilibrium was found to be $2.54 \times 10^{-3} \text{ mol}$.

Using the data above, calculate the following:

- the initial amount of $\text{Fe}_2\text{Cl}_6(\text{g})$;
- the amount of dissociated $\text{Fe}_2\text{Cl}_6(\text{g})$;
- and hence the degree of dissociation of Fe_2Cl_6 at 350°C .

$$\text{Degree of dissociation} = (\text{fraction of dissociated } \text{Fe}_2\text{Cl}_6(\text{g})) \times 100 \%$$

[4]

$$\text{Initial amount of } \text{Fe}_2\text{Cl}_6 = \frac{0.450}{55.8 \times 2 + 35.5 \times 6} = 0.001386 \text{ mol}$$

Let amount of Fe_2Cl_6 dissociated at equilibrium be $a \text{ mol}$

	$\text{Fe}_2\text{Cl}_6(\text{g})$	$\rightleftharpoons$	$2\text{FeCl}_3(\text{g})$
Initial amount / mol	0.001386		0
Change in amount / mol	$-a$		$+2a$
Equilibrium amount / mol	$0.001386 - a$		$2a$

$$\text{Total amount of gases at equilibrium} = 0.001386 - a + 2a = 0.00254$$

$$\text{Hence, } 0.001386 + a = 0.00254$$

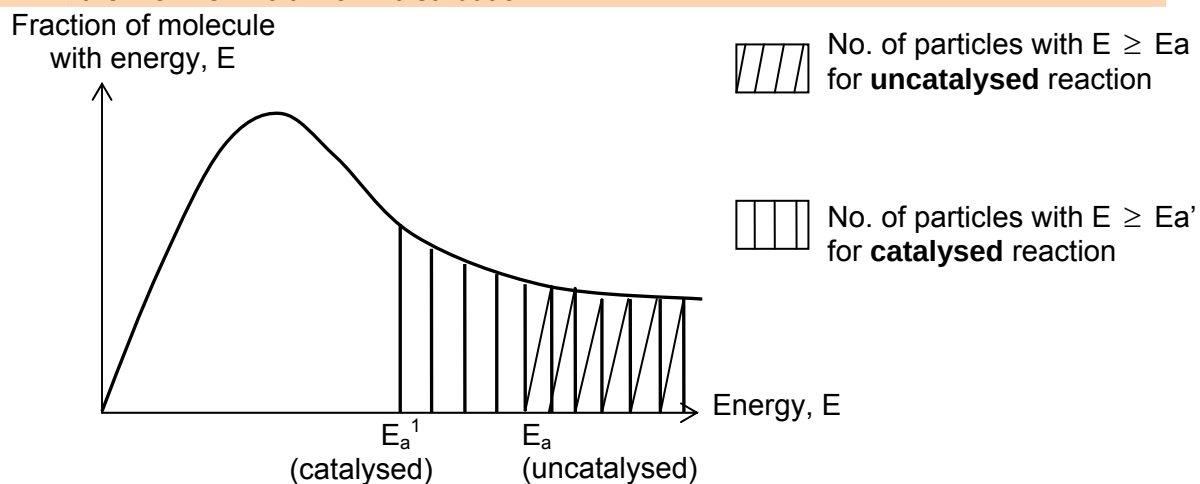
$$a = 0.001154$$

or [1] for any appropriate working to calculate the unknown value indicated by student.

$$\text{Degree of dissociation} = \frac{0.001154}{0.001386} \times 100\% = 83.3 \%$$

- (f) Iron(III) chloride is a catalyst used in the synthesis of chlorobenzene from benzene.

Explain the catalytic effect of iron(III) chloride on the rate of reaction in terms of the Maxwell-Boltzmann distribution.



When a catalyst is **present**,

- number of **reactant particles** with at least the **activation energy** ($E_{a'(\text{cat})}$) increases
- number of **effective collisions** taking place in the reaction **increases**
- rate of reaction is **proportional** to the **frequency of effective collisions**.
- **rate of reaction increases**

[3]